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Bringing human factors to life

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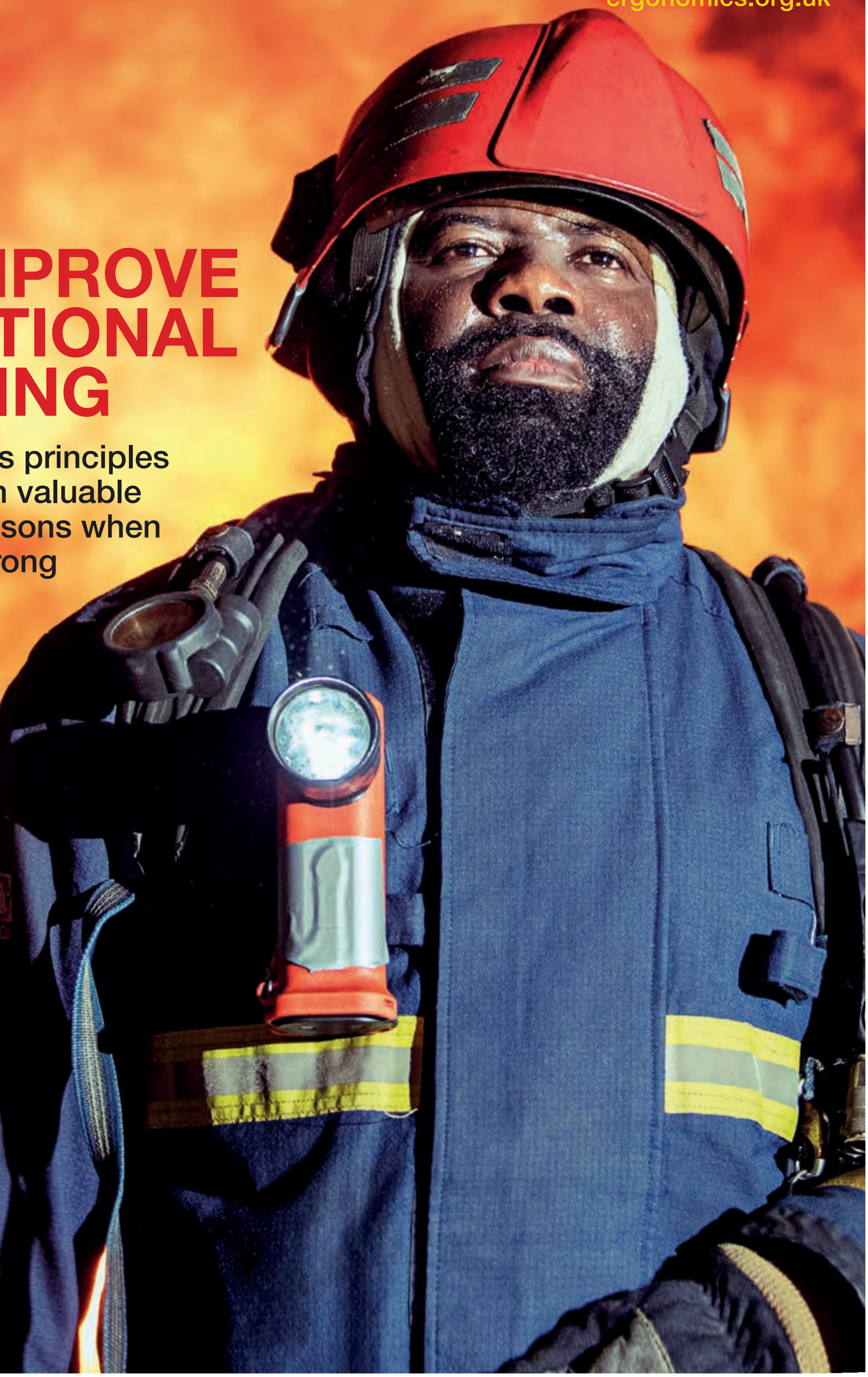

HOW TO IMPROVE ORGANISATIONAL LEARNING

Using human factors principles to ensure we learn valuable and sustainable lessons when things go wrong

Chartered Institute of Ergonomics & Human Factors

Late April 2021, London
conference.ergonomics.org.uk

Ergonomics & Human Factors 2021

Call for Papers

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- **The Future Workplace** including occupational health; cybersecurity; workspace design
- **The Built Environment** including teaching and education; social housing; complex use; smart cities
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PRESENTATION METHODS

You'll be able to present your topic in one of a number of ways:

- Live or pre-recorded talks
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- Podcasts
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All accepted papers will be published in the digital proceedings.

DEADLINES

Submission closes: 1 November 2020
Authors notified: 9 December 2020
Final submissions: 10 January 2021

How will you be involved?

For all details go to:
conference.ergonomics.org.uk

#ehf2021

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The magazine of The Chartered Institute of Ergonomics & Human Factors

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Chartered Institute of Ergonomics & Human Factors

The Chartered Institute of Ergonomics & Human Factors
Registered Address:
Edmund House
12-22 Newhall St
Birmingham B3 3AS
07736 893350

www.ergonomics.org.uk/think-green

EDITORIAL
Editor:
Tina Worthy
07736 893350
editor@ergonomics.org.uk

Editorial assistance
Connect Communications

Senior designer:
Gene Cornelius

Picture editors:
Akin Falope
Claire Echavarry

ADVERTISING
Sales Manager:
Daniel Goodwin
020 7880 6206
daniel.goodwin@redactive.co.uk

RECRUITMENT
ergonomist@redactive.co.uk
020 7324 2777

PRODUCTION
Aysha Miah-Edwards
020 7880 6241
aysha.miah@redactive.co.uk

ACCOUNT DIRECTOR
Will Hurrell
020 7880 8545
will.hurrell@redactive.co.uk

To subscribe to *The Ergonomist* please contact:
ergonomist-subscriptions@redactive.co.uk

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Redactive Publishing Ltd
Level 5
78 Chamber Street
London
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FROM THE PRESIDENT

Learning together

Many of us adopted new ways of connecting during lockdown. In June, we held our Council meeting via Zoom. I received Twitter training to support the Institute's social media activities. Beforehand, I didn't use Twitter and only had a handful of followers but very quickly, by engaging with other people's posts, my following grew and I learnt a lot about the work and interests of others. In my first week I even had a conversation about my Design for Everybody President's Project with Caroline Criado Perez, author of *Invisible Women*, with a 102.3K following! She said she'd love to see the project report when it's ready. We're running some great courses about how to get the most out of Twitter and LinkedIn and improve your online presence. If you missed them, you can still access the recordings.

We launched the Learning from Adverse Events white paper, written by an impressive team, led by Ron McLeod. I thought the launch webinar succinctly explained the nine principles for human factors in incident investigation and learning. The recording and white paper are both available to download. And we've had a great response to our Creating a Safe Workspace guidance and webinar with Workplace Sector Group Leads Kirsty Angerer and Ed Milnes. Keep an eye on our events website and social media to see what's coming up.

On behalf of the CIEHF, former President Neil Mansfield and I joined the Scientific Committee for the IEA 2021 conference. The call for papers is open until 25th September but if you're interested in reviewing papers instead, please get in touch.

It's great to see a buzz of activity around our Communities forum too. At the time of writing, John Lovegrove's discussion about the gap (or not) between practice and research is particularly popular. It's proving to be a useful platform for sharing methods and papers, as well as views. I hope we continue to work together in this way.

Amanda Widdowson
CIEHF President

president@ergonomics.org.uk

It's great to see a buzz of activity around our *Communities* forum

FROM THE EDITOR

Adapting to a shifting future

As Covid-19 continues to dominate our lives and the world around us, we start to look at how and what we can learn from what has happened, how we reacted in the immediate aftermath and ongoing efforts to control the situation. Mark Sujan introduces our new guidance that aims to help capture valuable learning in the health and social care sector, Sue Hignett reveals the results of a survey into the challenges of PPE use and Siobhan Burns tells us first-hand, about her experiences as a trainer in patient handling in the last few months.

But of course, it's not just the current pandemic that continues to provide opportunities to learn, as our cover article featuring our newly released white paper on 'Learning from Adverse Events' explains. The paper focuses on nine principles that guide organisations towards sustainable learning. Courtney Grant describes his own involvement in giving emergency aid to those affected by the Grenfell fire, and Jo White focuses on where more could be done to learn from incidents in aircraft maintenance activities.

Looking to the future, Richard Bye and Andy Hawkes give us views into how human factors practices in rail and consultancy, respectively, are effectively adapting to new ways of working. Suzy Broadbent gives us an insight into jet fighter cockpit design and Clint Howells introduces a new framework that will provide cutting edge research in defence and security.

We also have a piece about work morbidity, a view of ergonomics in Finland and stories from four more of our award winners.

Enjoy the rest of the summer.

Tina Worthy
editor@ergonomics.org.uk
@ciehf

Gowns, face and eye protection and gloves have been deemed essential to stop NHS and social care staff getting infected with Covid-19 but criticism about the supply, use and functioning of these safety-critical items has been persistent, as a recent survey highlights

Protecting workers from infection

IMAGE: GETTY

Healthcare workers have borne the brunt of the Covid-19 pandemic, and it's been a very tough battle for them. It has cost the lives of at least 540 frontline staff in England and Wales - a higher number than anywhere in the world apart from Russia. On top of this grim and sobering statistic, there's been controversy in the UK since the start of the outbreak about the supply and use of personal protective equipment (PPE).

At one point in April, the British Medical Association said doctors dealing with the virus were frightened and being left with difficult choices about whether they should risk their lives by treating patients because of a lack of appropriate equipment. In the inevitable future inquiry into the UK's response to the pandemic, the issue of PPE availability and suitability will doubtless be examined in detail. However, we now have an early indication of the sort of responses that will be presented for public scrutiny in this investigation.

An online study carried out by Loughborough University has been looking at the real-life experiences frontline healthcare staff treating Covid-19 patients have had with PPE. A team led by Sue Hignett, Professor of Healthcare Ergonomics and Patient Safety, used Twitter and LinkedIn to carry out a survey of those working in UK hospitals. The study received more than 400 responses when it ran from 4 April to 8 May - the peak of the epidemic, when there were more than 33,000 deaths.

The researchers wanted to gain a quick and preliminary understanding of the use of PPE for prolonged working. Rather than looking at issues over the supply of the necessary equipment or infection transmission, the survey concentrated on whether the kit was comfortable and effective and if it changed the way staff worked. Questions centred around the clinical areas respondents worked in, whether they knew what PPE they should be wearing, how well their face cover, apron, gown or gloves fitted, if there was any hurt or irritation, whether they could see to read, operate equipment or communicate, and their height.

Professor Hignett explains that the idea for the study came from a discussion with a colleague, Professor Jay Banerjee, who works as an emergency medicine consultant at a hospital in Leicester. He was having to wear PPE and was finding it difficult as it changed the way he worked.

"He felt we needed to explore this further. Staff in the NHS are used to wearing masks and aprons and so on, with a higher level of PPE for specific patients on an infection control ward, for example. However, the difference with Covid-19 is that once you're in the hot zone, you're wearing that kit all day. We're asking staff to work in quite complex PPE, and they haven't had the training to do that. Surgical

We were surprised by how strongly respondents wanted to recount their experiences

masks can fog up your glasses and you might have a visor on top. And some of the Public Health England recommendations include putting on an apron, then a surgical gown, and then possibly another apron on top of those. That's going to be hot, as it's a lot of layers to wear."

Alongside this discomfort, staff were also needing to put on masks that left marks on their faces. Their patterns of working often also changed, with shifts being extended from eight to 12 hours. "We wanted to find out how people's tasks changed. For instance, we knew that they were starting to have to wear double or even triple sets of gloves. If you're doing a delicate job that involves needles and other sharp instruments and you've lost some of your haptic feedback because of these extra layers, then it makes it harder. It's not something you've been trained to do."

The survey contained simple tick-box questions with some boxes left for additional comments. "We intended it to be very quick as it was to be a snapshot to find out what the problems were. One of the things we wanted to check was if there was a difference between men and women. People were raising the issue on social media that they couldn't get their masks to fit their faces; they were all too big."

Professor Hignett and her team decided to evaluate the issue in four different ways:

- **Wearing PPE (anthropometry).** How well the PPE physically fitted the survey respondents.
- **Working in PPE (task analysis).** Did PPE alter what clinicians were doing and if so, how? Did any changes relate to wearing it?
- **Surviving PPE (injuries).** Did the kit injure people such as through skin breakdown, marks on the face or musculoskeletal neck problems caused by holding their heads in a certain position in order to see.
- **Supplying PPE (supply systems).** What was actually available and did it match up with the PPE they'd actually been fit tested on? Just how chaotic was it?

Clinical staff wear personal protective equipment (PPE) while caring for a patient in the intensive care

"They couldn't always order the PPE they had been tested for; it might, for example, have come from a different manufacturer. These problems have probably eased a lot now but it looks as if at the time, the government was sending kit out to different hospital trusts that may or may not have been what they needed."

Of the responses received directly relating to Human Factors/Ergonomics (HFE), 72% were from women and 28% from men. Nearly half (49%) reported communication and hearing issues with wearing masks and visors, with women flagging this up significantly more than men. 23% complained of visual difficulties with safety glasses, and for visors, the figure was higher at 27%. The fit of the equipment was reported as a problem, too.

One respondent said: "Apparently masks for smaller faces don't exist." Another reply was: "I'm quite short and overweight and find that the all-in-one suits do not fit me very well due to the proportions."

There were, perhaps predictably, also negative comments about injuries and overheating: "Two plastic aprons and a surgical gown as per guidance can get very sweaty and hot doing manual work which can make it harder to focus or continue to put the same effort in. It contributes to stress and exhaustion."

The survey report concluded that PPE acted as a safety barrier by changing the task interface. It said: "We suggest that more HFE research is needed to improve the functional design of PPE so that healthcare workers are better supported to carry out critical care and other medical treatment."

Professor Hignett says she was surprised by how strongly respondents wanted to recount their experiences. "Normally when you do this sort →

That's why at CIEHF we're having a big push to show people what we can offer and how we can help."

One really important point that concerns Professor Hignett is the fact that these issues with PPE were allowed to arise at all. "This was all foreseeable," she says, "it should all have been tested and planned and people should have had training, in the same way you'd have fire training every year. It's particularly true in intensive care units and emergency departments, as they're right on the frontline and where people come in. They should have been trained to the point where they could put on the required level of PPE and feel that their clinical abilities hadn't changed."

She believes that the publication of the survey will make a valuable contribution to the debate. But what comes next? "We've put in a research application to the National Institute of Health Research on their Covid-19 call for rapid research work. We're now waiting to hear the outcome."

If successful, the grant will allow the university to work with University Hospitals of Leicester NHS Trust for 12 months, looking again at the four issues of wearing, working in, surviving and supplying PPE. "We'll have access to their intensive care units and they'll be a fantastic partner to work with. The aim is to redesign PPE so that it works. We want to create the design criteria around people's different shapes and sizes and that's particularly important for women. Really, this research should have been happening in healthcare for the last 30 or 40 years. This may be an opportunity to finally carry it out."

of a survey and you ask people for comments, you don't get many. But for this, we received more than 250. That's a huge number. People really wanted to tell us what was going on. And when you read through those comments, the pain would come through. People felt that they were trying really hard to do their jobs and look after their patients, but the interface and what they were being asked to wear – it was just raw emotion. I couldn't believe it was happening."

There were also comments about the gowns, with people remarking that they didn't properly cover them, so their levels of protection were compromised and they felt exposed to the dangers of the virus. "Some staff were so adversely affected by the PPE and suffered so much soreness" the Professor adds, "they wondered if they'd actually be able to wear face masks when they went back on their next shift."

"We were hearing that safety glasses were too big and kept slipping, that goggles applied pressure and that the area behind their ears was red and raw. People said they had been wearing FFP3 (the highest protection level) masks and their noses had become red and started to break down. I even heard anecdotally that people had actually decided not to wear the PPE, which is just dangerous."

More HFE research is needed to improve the functional design of PPE

The survey responses raise serious questions about the way in which human factors is integrated into the work of the NHS. "It's one of the big issues. HFE experts haven't been working within the healthcare sector at anything like the level of engagement of pretty much all the other safety-critical industries such as in energy or rail, where there's been a lot of focus on how people work. Many of the tasks and activities people carry out in healthcare have never had an HFE assessment.

Medical staff transfer a patient

Sue Hignett is Professor of Healthcare Ergonomics and Patient Safety at Loughborough University.

Further reading

A paper on the survey conducted by Professor Hignett and her colleagues, *Human factors issues of working in personal protective equipment during the COVID-19 pandemic*, appears in Anaesthesia 2020, published by the Association of Anaesthetists. See https://onlinelibrary.wiley.com/doi/full/10.1111/anae.15198

IMAGE: GETTY

CHIEF EXECUTIVE'S PERSPECTIVE

Human-centred design and the next wave

The Achieving Sustainable Change guide explains the two key areas in achieving effective organisational learning: mindset and action

The CIEHF has without doubt made, and continues to make, a substantial contribution to the response to COVID-19. Professor Mark Sujan's guide on *Achieving sustainable change: Capturing lessons from COVID-19* is the latest in a series of documents we've published. The guide, available free from **https://bit.ly/HFSustainableChange**, provides templates for capturing learning that can be fed back into an organisation using systems thinking. An infographic to accompany the guide can be found at: **https://bit.ly/HFSustainableChangeInfographic.**

Our work has been reported widely in the national press, the leading general medical journal the *BMJ*, and celebrated by the International Ergonomics Association and other societies. Our members are being invited to speak on national and international platforms reflecting the impact we're making as the voice of the discipline and profession. As I write, we're arranging for speakers to address the UK Parliamentary and Scientific Committee on the topic "How human-centred design can make a difference to future crises".

We're all poised for what might happen next and when a likely second wave may occur. This backcloth is further exacerbated with other world issues from Black Lives Matter and the search for equality to the impact on industry and commerce that will lead to a recession. At the same time, technology firms continue to bring innovations to the market that will change the way we live and work forever. Our Special Interest Group on Digital Health and AI will lead research and debates in this area.

Working with PARN, the Professional Associations Research Network, the Institute will put in place a plan to ensure that we are a representative body and that diversity and inclusion is core to our work. After all, human-centred design is for everybody. The President's Project, *Design for Everybody*, is a good example of how we're being proactive. See **https://bit.ly/DesignForEverybodyGuidance.**

The CIEHF has made a substantial contribution to the response to COVID-19

Right now, we're looking for international case studies that demonstrate how human factors can make a difference to business. These will be promoted to businesses throughout the UK to reinforce our role in an economic recovery from outstanding designs that contribute to revenue generation to new and different ways of working that will add value. Case studies or enquiries can be sent to emmacrumpton@yahoo.co.uk.

We've also launched training to help improve members' awareness and practical skills in using Twitter and LinkedIn. The purpose of this is to help members better promote themselves, their work and of course our discipline and profession. We're looking for individuals who would like to receive such training and coaching helping them to become Social Media Champions for the Institute. Anyone interested should contact me directly.

Noorzaman Rashid
Chief Executive of the CIEHF
Noorzaman.rashid@ergonomics.org.uk

noorzaman rashid

IMAGES: GETTY

BAE Systems, Barrow-in-Furness

Building a secure future

With an invitation for organisations of all shapes and sizes to get involved, **Clint Howells** introduces the Human Social Science Research Capability framework, and explains how it will provide cutting edge research to address some of the UK's highest priority human science issues, helping to shape future defence and security strategy, policy and capability

In March 2019 the Human Social Science Research Capability (HSSRC) framework was contracted, with BAE Systems as its Prime Contractor. The framework enables human, behavioural and social science research to be commissioned by the Defence Science and Technology Laboratory (Dstl), the Ministry of Defence and other UK government departments including the Home Office, Departments for Education, Transport, International Trade and Work & Pensions, the Ministry of Justice and the College of Policing.

Cutting-edge research related to defence and security is commissioned across six technical themes:

- Personnel
- Training and education
- Humans in systems
- Human performance (physical and psychological)
- Understanding and influencing human behaviour
- Health, wellbeing and enhancing medical systems and capabilities

The future environment will present different physical and psychological demands which need to be identified, understood and managed. HSSRC-commissioned research will find innovative ways to attract and retain skilled people, maximise human performance and effectively integrate and exploit automation. The HSSRC also offers capabilities to analyse human behaviour and communication, including the use of social networks, which will be increasingly important to support understanding of the future information space.

The HSSRC framework can be used to commission short-term and multi-year research tasks, as well as facilitated workshops, sandpits, open calls and provision of expert advice. Outputs from the tasks are likely to include reports, data, prototypes, software, presentations, educational materials and recommendations. Some recent tasks include:

- Study of new techniques to improve cognitive performance under operational conditions.
- Development and evaluation of models of skill retention and decay.
- Research into the effects of human-machine interface design on training and human-system performance.
- Creation of a non-invasive cognitive workload assessment tool for use with individuals or teams in the field.
- Investigation into how society, technology and the nature of work are changing, to help defence adapt and reflect the workforce in 2030+.
- Review of literature and technologies relating to the use of exoskeletons to improve performance.
- Design of interventions and measures to enhance teamworking in newly formed teams.
- Exploration of ethical and philosophical worldviews relating to cyber operations, and how these can influence command and control and decision making.

The future environment will present different physical and psychological demands which need to be identified, understood and managed

The framework operates through a network of suppliers ranging from micro businesses to large organisations and universities. The vast majority of tasks are open to all signed-up suppliers who can view requirements and participate in competitions alone or in collaboration with others. New suppliers can join the framework at any time and the HSSRC welcomes interest from companies which have not previously worked in defence or security, and who can provide fresh perspectives from other domains.

IMAGE: BAE SYSTEMS

Suppliers contributing to this important work will help to shape defence and security thinking, policy and doctrine, underpin procurement decisions and improve the UK's operational capability. They will also have the opportunity to attend supplier events, develop links with other members of the supplier network and propose innovative, cross-disciplinary solutions to challenging and exciting research requirements.

The framework launched its first requirements in June 2019 and will run until March 2023 with a potential extension to 2026. Multiple defence and security-related tasks will be let each year, ranging from £20k to £1m+. The HSSRC is managed by a small team based at BAE Systems' office in Yeovil, Somerset. The team is supported by a group of experienced Research Theme Leads who provide technical assurance and ensure coherence between the research tasks. A pool of Technical Specialists is available to provide additional support and niche skills.

The HSSRC is led by Dr Helen Cole. Helen was a member of the Human Factors Research Group at the University of Nottingham and led research activities examining usability and user-centred design of novel technologies in educational settings. She worked as a human factors engineer in the defence industry before moving into project and programme management roles with a strong focus on leading research and technology programmes.

Dstl's HSSRC Strategic Lead Jim Squire commented, "In this time of crisis, as the whole world is united in its fight against the coronavirus, Human and Social Sciences have never been more important in supporting the defence, security and prosperity of the UK. It is critical that the MOD is able to work with the very best talent in academia and industry to address people-related opportunities and threats, and I have no doubt that the HSSRC framework will deliver high impact Human and Social Sciences research for years to come."

How to get involved

If you would like more information on the HSSRC or are interested in joining the supplier network, please get in touch with the team at hssrc@baesystems.com. We are also keen to identify additional technical specialists who can provide expert support to tasks on a flexible basis.

Clint Howells MSc C.ErgHF MCIEHF is an independent consultant with a background in human factors integration for large engineering projects. His current interests include future platform usability and strategic supply chain development for the HSSRC.

A no-touch future

An article on a website called strategy+ business delves into the urgent and growing need for contactless technology at a time when the coronavirus has spread around the globe with frightening speed.

The author explains how speech recognition, facial recognition and digital money are enabling us to move to a world where multiple people don't have to handle and pass between them physical items such as passports, menus and credit cards. Instead technology will allow us to reduce the risk of virus transmission by allowing us to say what we want instead of using a public touchscreen or pressing a much-touched button, for example. Millions of people are already asking things of Alexa, after all, so will the technology soon be used in public places?

Many are already comfortable with allowing their laptop or mobile device to recognise their face or fingerprint and activate, and airports are using facial recognition for passport checks. Perhaps it won't be long before our cars and houses will unlock when they recognise us approaching.

Contactless payments are becoming much more common and in many cases are already a necessity where retailers will not handle cash due to the potential contamination risk that comes with it.

For more details, see **www.strategy-business.com/blog/Technology-for-a-no-touch-world.**

An ergonomics perspective from Finland

In 1906, Finland became the first European state to grant all adult citizens the right to vote, and the first in the world to give all adult citizens the right to run for public office. Today, with its population of just 5.5 million, Finland ranks highly in quality of working life, as Päivi Steffen and Arto Reiman explain

Since its independence following the 1917 Russian Revolution, Finland has been known for its competitiveness in manufacturing, focusing first on basic industries like wood processing and metals and, in recent years, on telecommunications and electronics. The geographical position of Finland between two very different countries, Russia (the former Soviet Union) and Sweden, has significantly impacted the development of the Finnish economy. Finland has benefitted from trade with Eastern Europe but, as a part of the European Union, the country also engages in global business and has been among the best performing economies in the European Union.

Over recent decades, Finland has benefitted too from the success of Nokia and its subcontracting network but since the last recession in the 2010s, the role of Nokia has decreased. Meanwhile, a number of promising start-ups in information and communications technology, gaming, cleantech and biotechnology have been created but the largest public and private employment sector in Finland is the health and social and care sector. In recent years, there has been frequent discussion of the economics of this sector and the need for the reform of the system.

The Finnish economy recovered quite slowly from the recession and, in 2016, the government enacted a competitiveness pact aimed at reducing labour costs, increasing hours worked and introducing more flexibility into the wage bargaining system. The aims of this pact were only partially achieved and now the country, among many others, has suffered from the spread of the coronavirus. It's highly likely that Finland will continue to seek new ways to promote its future competitiveness although the general trend of an ageing population and workforce places further demands on the economy.

Various indicators show the long-term success of Finland when it comes to the quality of working life. For instance, the statistics of the European Foundation for the Improvement of Living and Working Conditions (Eurofound) show that the quality of working life in Finland ranks among the highest in Europe. The statistics of the European Agency for Safety and Health at Work indicate Finland's successes in occupational safety and health. From an ergonomics perspective, working life in Finland has changed significantly during the last few decades. Heavy, manual work has decreased, while, in many ways, the cognitive load at work has increased. Nevertheless, musculoskeletal disorders remain a high contributor to sick leave and premature retirement from work.

Finland is known for its unique occupational healthcare system within which, in principle, employers are required to provide occupational healthcare services to their employees. A common practice has been to source the services from a commercial partner, that is, from an occupational healthcare provider which has led to the success of the private occupational healthcare sector in Finland.

The role of the occupational healthcare provider has been to provide both preventative healthcare and medical care, with the latter being the more dominant service. In many ways, this arrangement has affected directly and indirectly the role of ergonomics in Finland. Traditionally, ergonomics has been considered as part of the Finnish occupational healthcare system but again the focus has been on medical care, leaving little scope for preventative ergonomics. However, in

Aurora Borealis seen in Tampere, Finland.

2018, a regulatory change modifying the national compensation system meant that the role of the occupational healthcare provider has shifted towards more preventative healthcare services.

It remains to be seen whether this process of change will initiate new openings for ergonomics. Ergonomics in Finland is, in a sense, at an intermediate stage where certain key actors or process owners are missing. At the workplace level, ergonomics is considered a service that's provided by the occupational healthcare provider but the services offered are general and rather limited when it comes to the broad spectrum of ergonomics. Also, other potentially relevant actors, such as occupational safety and human resources specialists, often lack a basic knowledge of ergonomics which hinders possibilities for company-oriented ergonomics development processes. In addition, the definition of ergonomics is, in many ways, unclear in workplaces resulting in a very low number of ergonomics consultants and European certified ergonomists in Finland.

Ergonomics education at degree level is provided at only one university, the University of Eastern Finland, where ergonomics is a major programme under the Faculty of Health Sciences but in Finnish technical faculties there are no direct ergonomics professorships or graduate programmes. This is somewhat surprising considering the design-oriented nature of ergonomics that calls for multidisciplinary approaches, including engineering, for the understanding of human factors in product and overall system design. This interpretation of the current state of ergonomics at the university level might still be too simplistic. Technical universities teach various topics that could be numbered among the broad concepts of ergonomics. At the University of Oulu, for instance, occupational safety and health, wellbeing at work, and productivity and sustainable organisations are taught to engineering students. The University of Tampere runs a master's degree programme in safety and security. So, although ergonomics as a discipline might be missing from the curriculum of technical faculties, the core principles of ergonomics system design and the dual optimisation of wellbeing and productivity are being taught to engineering students.

In Finland, ergonomics has been and still is challenged in many ways but there are certain signs that could be interpreted as indications that the role of ergonomics might be about to change. For instance, the role change for occupational healthcare providers towards preventative healthcare might open new possibilities for ergonomics services. Also, the government programme aims to promote a socially sustainable working life which has potential and possibilities for ergonomists in Finland.

Heavy, manual work has decreased, while the cognitive load at work has increased

A dentist and her assistant treats a patient at a practice in Helsinki.

Päivi Steffen MSc works as an executive manager in the sports sector and is a member of the board of the Finnish Ergonomics Society. Her interests lie in organisational and cognitive ergonomics.

Adjunct professor **Arto Reiman** leads a research team focusing on wellbeing at work and productivity at the University of Oulu in Finland. He is the chair of the Finnish Ergonomics Society.

Further reading

Eurofound: www.eurofound.europa.eu/country/finland#actors-and-institutions

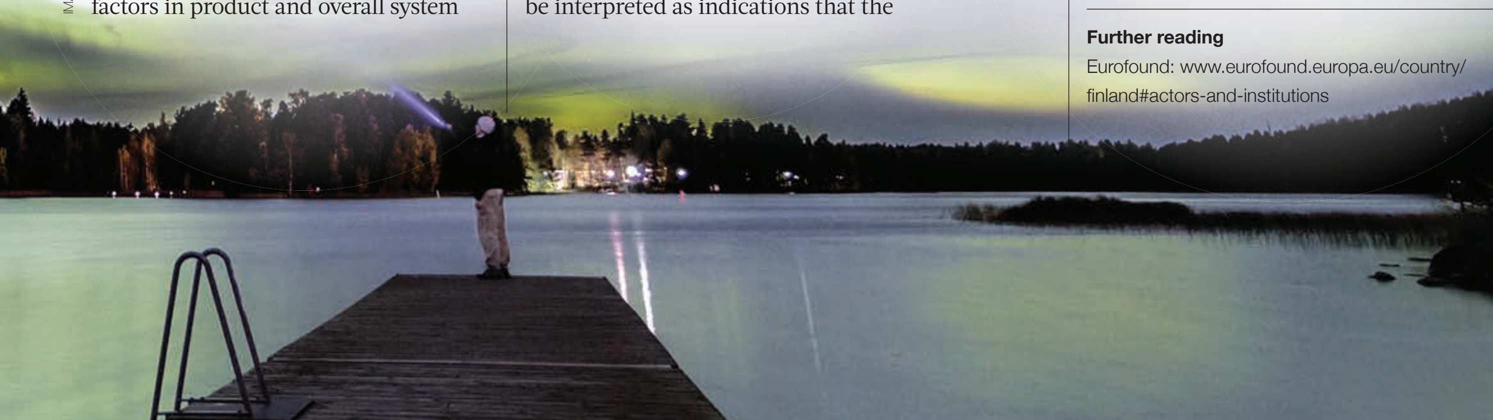


IMAGE: GETTY / ALAMY

Learning from COVID-19

As part of the response to COVID-19, CIEHF developed a new guide to help capture some of the valuable learning created as staff and organisations in health and social care adapt to the challenges posed by the pandemic. Mark Sujan and Jess Wadsworth and colleagues explain how the guidance can help to achieve sustainable change

We know that learning lessons and putting them into meaningful change is a tremendous challenge for the NHS. In part, this is due to the well-intentioned but somewhat narrow focus on learning from incidents. This perspective has been too limiting, and often organisational learning has been reduced in practice to an administrative exercise of counting frequently occurring adverse events, such as patient falls.

With this new guide, *Achieving Sustainable Change*, we drew on insights from resilient healthcare and complex adaptive systems. We want to encourage people to think about how they navigate successfully difficult situations that are full of uncertainty and where there is always a lack of resources and seemingly endless demand. Looking at how people anticipate changes, how they monitor situations and how they adapt the way they work to the specific context, can help organisations to become more resilient.

The guidance provides prompts for organisations to reflect on what goes well even when situations are challenging, and how safe spaces can be created where staff can contribute to organisational learning, and where they can take ownership of improvement and change.

The red zone

Epsom and St Helier University Hospitals NHS Trust in South London was one of the first to experience the significant rise in COVID-19 patients requiring critical care (CC) and quickly enacted its pandemic plan to open a large, temporary CC unit at Epsom. This was an enormous feat as equipment and resources were pulled from across sites, staff were redeployed from multiple settings including community, and the new unit was open within weeks.

Suddenly there was a group of staff, of different professions, specialities and levels of experience, thrown together by circumstance. The redeployed doctors, nurses, physiotherapists, psychologists, healthcare assistants, administration staff, yoga teachers and project managers came from far and wide to help. Some took on roles in staff wellbeing, relative liaison, PPE support and behind-the-scenes organisational management, particularly if advised against working in red zones (the highest COVID-19 risk areas).

However, inside the red zones it felt like a different world, like a battle where nothing stayed still. All staff were now faced with a situation where they were being asked to work completely out of their comfort zone and, in some cases, scope of practice. Many redeployed staff had never set foot in a CC unit, let alone looked after an intubated, ventilated COVID-19 patient, and had not experienced death on this level. Staff were scared, anxious and concerned about the blame culture that still prevails in healthcare. Even the most experienced CC staff had not seen this volume of sick patients at any one time, often with multi-organ failure, requiring a fine balancing act of ventilation and prone positioning to improve oxygenation, continuous inotropic drug infusions to support blood pressure and haemofiltration to replace reduced kidney function. Instead, CC nurses who normally look after one patient were being asked by NHS England to look after up to six, with the inability to give the normal standards of care causing all staff some moral injury.

Initially the simulation and human factors team became involved with the temporary CC unit to deliver just-in-time simulation training for redeployed staff. However, it soon became apparent from observing work-as-done and listening to staff feedback that we could use both our multi-professional clinical expertise and human factors perspective to work collaboratively with the senior CC team. Together we designed several systems and processes that could help staff to work better in this complex, emergent situation. This included three things. One was the development of new COVID-19 adapted staff role expectations and responsibilities to allow a safer and more equitable allocation of skills, resources and workload. Another was the streamlining of handover processes and allocations to encourage a more multi-professional and collaborative approach. The third was the introduction of a coloured hat system so critical care skills could be easily identified to improve communication and escalation processes. We also offered daily orientation and basic training for new starters and adapted the standards of nursing care to ensure the workload was realistic and achievable.

Staff could clearly feel the benefits of these interventions as the unit felt calmer and more organised and the clearer expectations for each role made staff feel they "knew what they were supposed to do", reducing cognitive load. However, this adaptation did not work as well during shifts when staffing levels were lower or where there was resistance to the new processes from some team members. This meant both bedside practitioners and CC nurses were focused on giving direct patient care and left a reduced number of CC staff with oversight of the bedside practitioners and department. These are some of the important lessons we will need to consider in order to cope with a second wave.

IMAGE: GETTY

Available to download: *Achieving Sustainable Change: capturing lessons from COVID-19*

Organisational learning

We know we must learn from COVID-19 but how do we decipher the learning from such a complex and immediate situation and make it valuable? The temporary COVID-19 CC unit had closed by the time this new guidance was published but using the template retrospectively really helped us to organise events, guide reflections and illuminate key learning points from everyday work to take forward. Without going through this process, the events and interventions entwined within the COVID-19 chaos were likely to have gone unnoticed. They have now been captured, informing how we organise ourselves in the event of a second wave.

Learning the 'right things' is a problem in healthcare as we are overwhelmingly persuaded by our risk and governance systems to focus on what went 'wrong'. This is often addressed by a series of actions involving training or a revised policy, which seldom moves us forward. Working through the template took time and a level of understanding around the mindset and fundamental principles of systems thinking which in reality, people working on the frontline of healthcare may not have. Yet this process of understanding what happened and why has raised our gaze to systems level interventions that can be proactively utilised in the future, rather than a traditional retrospective style investigation of looking at what went wrong, which is far more likely to focus on individuals.

So how can we adopt this guidance into our safety toolbox? A huge amount of time and resource is afforded to incident investigation and quality improvement, so why not try to complete the template alongside a traditional report or before the implementation of a quality intervention? We also need to think smartly about the skills and resources already in our organisations; who are the people or departments that can take this template, use it to collaborate with those doing the work, and put the learning into practice? When we look through the lens of organisational learning and systems thinking, only then can we truly begin to understand what is happening in our organisation and use this information to strengthen and grow.

Dr Mark Sujan is founder of Human Factors Everywhere, a small company dedicated to helping people adopt human-centred approaches for running a safe and secure business. He leads the CIEHF's Digital Health & AI Special Interest Group.

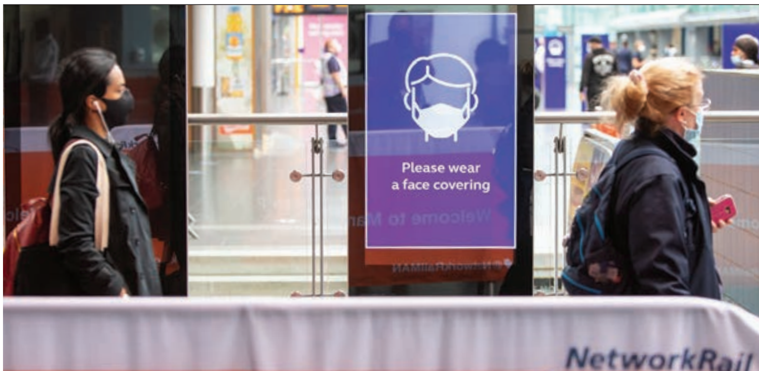
Jess Wadsworth is a Senior Nurse Educator and the Deputy Lead for Simulation and Human Factors. She is a member of the Civility Saves Lives group and has an interest in inter-professional education and human factors.

Dr Sabrina Vitello is a Senior Simulation Fellow with a clinical focus in palliative medicine and a particular interest in exploring human factors.

Dr Jennifer Blair is a Consultant Anaesthetist and the Director of the Centre for Simulation and Human Factors. She's passionate about the practical application of human factors to healthcare.

Further reading

Download *Achieving Sustainable Change: capturing lessons from COVID-19* from https://bit.ly/HFSustainableChange



Work as (re)imagined

Overpriced sandwiches, queuing at the coffee machine, commuting and team meetings, are just some of the things **Richard Bye** never thought he'd miss, until, that is, lockdown reared its rather peculiar and totally unexpected head

As a member of Network Rail's Ergonomics team, I'm all too aware how lucky I am to have a pandemic-proof role. But although the team's jobs are secure, none of us have been immune to the effects of enforced home working. We entered lockdown with a sense of pragmatism; thinking through our collective risks, assumptions, issues, dependencies and opportunities, so that we could plan how to maintain the essential contribution that ergonomics and human factors work makes to the safe and effective running of the GB rail network.

What followed were several long days, as we tweaked tried and trusted techniques, and examined new and innovative methods, to ensure that work we were having to re-imagine would translate into work as it needed to be done. This necessary questioning of what we do, and how and why we do it, was an interesting experience and a valuable exercise in introspection. It highlighted the need for a systematic approach (based on good practice ergonomics and human factors methods) to understand the team's goals, scenarios and task demands following the highly volatile step-change in our operational context of use.

No/go-to methods

When lockdown was thrust upon us, some of our go-to approaches for improving human-centred performance were immediately rendered impossible, and even though we needed one very quickly, we had no plan B (or C, D or E). The need for contingency to cope with the complexities and uncertainty of a global pandemic brought to the fore the realisation that agility needs to be a state of mind as well as a practice.

To protect the vital staff who keep the railway running, a range of practical measures (such as the use of Perspex screens, PPE and new procedures) have been put in place. And crucially, access to frontline locations has been severely limited. After all, with just one ill-advised trip to a control centre, filled with people possessing specialist skills, expertise and local knowledge, someone could unwittingly pass on COVID-19 and in doing so severely limit the capacity of a large area of the rail network.

As such, our day-to-day and face-to-face interactions with signallers, controllers and maintainers have had to be put on hold, along with the hands-on activities that are necessary to inform changes to the design of work systems. Nonetheless, work still needs to be done to manage the human factors considerations emerging from perturbations in the socio-technical system that is the GB railway. To this end, a range of novel workshop, meeting and presentation technologies have been employed to deliver the analysis, design and evaluation interventions required.

Stationed at home

Four months in and a return to the office still seems some time away. Having spent over a hundred working days navigating our way through a situation none of us ever imagined being in, we now don't know what to think. Is this really our new normal? What is normal anyway? And in any case, given that pre-lockdown we often worked remotely with a heavy reliance on technology, how new is all of this in reality?

In practical terms lockdown working has been a success, albeit quite hard going at times. Ergonomics and human factors activity on all major rail projects has been able to continue in some shape or form, thanks to the dedication and diligence of the team and the numerous CIEHF Registered Consultancies and Chartered members who work alongside us. Indeed, it's a testament to the collective skills and experience of all the human factors professionals involved that there have been no significant delays or problems in the ongoing design and delivery of improvements to the rail network.

We tried to ensure that work we re-imagined would translate into work as it needed to be done

Next stop

We're currently in the process of a retrospective analysis of the team's response to lockdown working. Resilience engineering research prioritises the need for continuous learning so we need to understand the effectiveness and sustainability of any new ways of working. In doing so, we've begun a long-term process to redouble our efforts to build and maintain emotional resilience, interpersonal support and shared adaptations in the face of global chronic unease.

Richard Bye is a Principal Ergonomics Specialist at Network Rail.

Getting to grips with new technology to help enhance ergonomics services

Making virtual consulting a reality

The COVID-19 crisis has seen digital businesses thrive, with the likes of Zoom seeing phenomenal growth that's likely to be sustained. **Andy Hawkes** talks us through how consultants can adapt their business practice to keep up with the changing times

No one can predict how the business world will behave in 2021 and beyond. Let's face it, we probably can't even predict what the world will be like next month! What is certain is that traditional face-to-face businesses such as consultancies will need to embrace every aspect of connected technology and software to ensure that the face-to-face element is no longer the main competitive advantage. That's not to say that expertise and experience are redundant or that some elements of face-to-face contact will still remain but the future will see an acceleration of virtual relationships with clients.

Video conferencing has been around for 30 years with a slow and gradual usage, yet, within three months we have taken to it in our business and social lives without any issues. Even the older generation who have been brought up with and have the engrained belief that "face-to-face is the only way to do a deal" mentality have realised that this was a false belief. And, of course Millennials and Generation Z just see the virtual world as the norm.

Digitise now and find partners who can help

Consultancies will need to look at their proposition and look at how they can virtualise and digitise products and services. Clients will expect their trusted advisors to be able to flex their approach to fit into the new way of working; they will still want the expertise but the delivery will need to change.

It's likely that many firms will see employees remaining as remote workers whereas others will be looking at how they can operate while COVID-19 remains a threat. This is likely to mean on-site visits from consultants, sales executives, suppliers and partners will be limited at best.

Be the early adopter and offer your clients and prospects as many alternatives, even if you don't have the alternative ready to hand. We see clients who want us to find solutions for them currently and we are willingly sourcing them, even if the alternative reduces our revenue and margin. I would rather have a happy client long-term than no client.

Technology adoption is increasing rapidly

In the ergonomics space, we're seeing a significant demand for, and a move towards, software and the virtual delivery of assessments. Many major global organisations want these solutions yesterday! The use of e-learning, online self-risk assessments, cameras, AI and sensors will see accelerated usage, but behind the scenes there's still a need for expertise to help design the solutions and to virtually deliver the actions needed to mitigate risk and pain. In every threat comes opportunity and this crisis will see creative, market-driven individuals and businesses responding to the new demands and risks that have already emerged, and will emerge, in the future.

Look to your customers: what do they need?

It's vital that consultants talk to their clients and find out how their expertise can be delivered in the virtual world. Just because you have had a face-to-face traditional relationship doesn't mean that a virtual, digitised version will not be acceptable.

If you need to invent a new solution don't be afraid to talk to potential partners about how expertise and software can combine and create new solutions to meet new working practices. Become experts in the use of webinars and video conferencing and, if needed, get some training in the use of the brilliant tools that exist. I'm sure that the better trained you are in virtual technology, the better you'll appear to your clients and prospects.

So, look forward to the second half of 2020 and into 2021. Use this time to rethink your business, products and solutions and grasp the new opportunities that will undoubtedly present themselves.

Andy Hawkes is CEO of Cardinus and has worked in the insurance and risk management sectors for 30 years and is immediate Past President of IIRSM. He's written widely on business risk and insurance risk issues and has specific expertise in speciality commercial insurance as well as compliance and governance risk.ergonomics.

IMAGE: GETTY

By not taking human factors into account effectively after they experience incidents or other potentially serious adverse events, many organisations miss the opportunity to learn lessons and implement effective change to mitigate and avoid similar events in the future, as a new publication from the CIEHF explains

How to improve organisational learning

Every day, so the saying goes, is a school day. It's an excellent phrase, and it makes a lot of sense. Our experiences shape us, so by the time we go to bed each evening, we should be that little bit more experienced and wiser. The only problem is that in the case of learning from adverse events, it may not actually be true.

According to many leading human factors specialists, the truth is that despite all appearances to the contrary, and despite the best efforts of so many well-intentioned people, we might not be absorbing the lessons we need to learn in order to stop them happening again.

CIEHF's newly published white paper into this problem, *Learning from Adverse Events*, based on three years of work from senior professionals, examines how and why organisations fail to learn fully after adverse events occur. It offers invaluable guidance on how to improve learning and implement sustainable change through increased understanding of the human contribution to those events.

The key to turning things around and achieving success, it says, is effective learning. This means "identifying and implementing change in a way that supports and sustains long-term improvement". The paper is based around nine principles for how to incorporate human factors into better learning from investigations. Taken together, they provide a comprehensive blueprint for improved performance.

THE NINE PRINCIPLES FOR LEARNING FROM ADVERSE EVENTS

1. Be prepared to accept a broad range of types and standards of evidence.
2. Seek opportunities for learning beyond actual loss events.
3. Avoid searching for blame.
4. Adopt a systems approach.
5. Identify and understand both the situational and contextual factors associated with the event.
6. Recognise the potential for difference between the way the work is imagined and the way work is actually done.
7. Accept that learning means changing.
8. Understand that learning will only be enduring if change is embedded in a culture of learning and continuous improvement.
9. Don't confuse recommendations with solutions.

Apparently simple incidents often have a multitude of interlinked contributing factors from which organisations can learn

IMAGE: PA

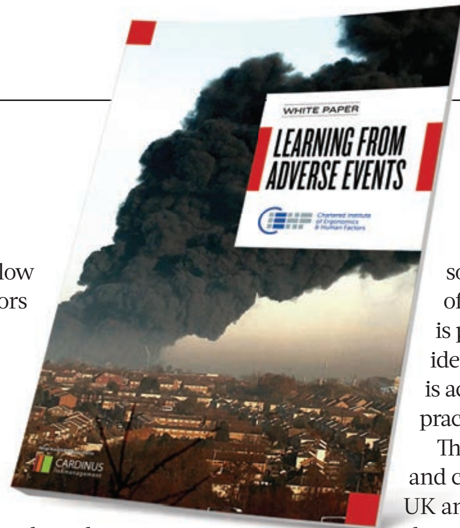
The lead author of the paper is CIEHF Chartered Fellow Professor Ron McLeod, an independent human factors specialist with almost 40 years' experience. As well as Ron, the core working group comprised Jon Berman, Claire Dickinson and Donna Forsyth. Ron says the paper is not aimed primarily at human factors professionals, but at those dealing with risk assessment, learning and improvement across business, industry and commerce who do not have a professional background in human factors. It was striking, he adds, how many CIEHF members wanted to contribute to the paper. "There were a number of themes that kept coming up. One was the reflex to find someone to blame – most obvious in the media – at the expense of understanding the systemic issues behind most complex events. Another was widespread concern about the continuing over-reliance on hindsight rather than trying to understand the situation and context at the time that individuals made decisions and acted."

Ron says that there was also concern among members about the widespread lack of awareness of the deep psychological motivations behind the way people behave, including the way they are incentivised and their goals and priorities. "Even when investigations were conducted well, there was a difficulty in coming up with recommendations that lead to genuine learning and improvement."

An adverse incident, he explains, can be of various types. It could be an event leading to actual loss or harm, or a near miss where nothing goes wrong but there's potential for serious failure. There's also the situation of what are often referred to as 'weak signals', where something is not operating in the way intended, nothing goes wrong but there is a potential for failure.

"It took a lot of work and some fairly brutal editing to get the white paper down to just nine principles," Ron says. "However, we now feel they capture the essence of what organisations need to know to improve. They're not in any order of priority but the principles should fairly readily lend themselves to self-assessment. Organisations should do a gap analysis and then implement the change." Few of the principles will be new to CIEHF members; they embody much of how most human factors professionals go about their professional lives nowadays. Ron says, "We were impressed by how often our contributors told us that, while the principles are well known to us as professionals, they are certainly not well known or applied by many people involved in investigations and learning, that need to understand and use them".

Professor Paul Bowie, Programme Director of Safety and Improvement at NHS Education for Scotland, spoke at a webinar launching the white paper. He said it contained a lot of information the health service could use. "These principles are particularly timely and highly relevant for what's happening in health and social care," he said. "It's unfortunate that in the significant minority of cases there is still so much of a blame culture; the way we report in the media around patient safety incidents, for example. The idea of adopting a systems approach often means something different in healthcare. There is a lot of work to be done there and so the guidance is particularly helpful in that respect. The whole idea of 'work as imagined' versus the way work is actually done is not well known, understood or practised across the NHS."



The CIEHF's latest white paper, a landmark publication

The white paper is now being widely promoted and circulated among risk managers both in the UK and around the world. For example, the current edition of *The Sentinel*, the magazine of the 8000+ membership of the Chartered Institute of Risk and Safety Management, has several pages highlighting the importance of the document and giving readers an extract of its content.

CIEHF Chief Executive, Noorzaman Rashid, says that the document has an absolutely central role in how incidents are investigated and learned from in the future. It is, he says, an outstanding example of how human factors specialists are contributing their expertise to a huge range of subjects across the business, industrial, medical and social fabric of life. "Rarely has a white paper been given this level of detail and attention to research making it a landmark publication and must-read document. It tackles everything from the blame culture – which does nothing to support effective learning from incidents – to differentiating between recommendations and solutions.

The white paper offers invaluable guidance on how to improve responses and actions through increased understanding of the human contribution to adverse events

"As Ron McLeod has said, many of our experts were ready and willing to contribute to this outstanding piece of work. I would like to take this opportunity to thank each and every one of them for their time and effort in making this document the remarkable success that it is. I'm absolutely certain that it will become a bedrock for the way in which incidents are investigated and learned from in the future," he says.

The white paper shows just how successfully the Institute is now working alongside allied professional bodies to ensure that the way problems are challenged reflects a systems-based approach that's so important to achieving sustainable positive change. "You only have to look to our recent COVID-19 guides and manuals to see how our experts, working with others from different bodies, are contributing knowledge at the highest level; knowledge that is helping to deal with incredibly difficult situations."

To access the white paper, please go to https://www.ergonomics.org.uk/CIEHFLearningfromAdverseEvents

Ron McLeod, lead author of *Learning from Adverse Events*

Further reading

To listen to the webinar which launched the white paper, go to https://events.ergonomics.org.uk/event/learning-from-adverse-events/

Mission-critical cutting edge technology

Being a military jet pilot is one of the most exciting - and stressful - jobs on earth. When flying at more than twice the speed of sound, decisions made in tiny fractions of a second can mean the difference between life and death but human factors can help ensure the right decisions get made

A Royal Air Force Typhoon pilot enters the cockpit

IMAGE: CROWN COPYRIGHT

The g-forces in flight can be massive and adrenaline levels go through the roof. And while all this is happening, an enemy may be doing their best to shoot you down but as a pilot, you're not alone. Modern aircraft have the very latest cutting edge technology to keep you safe and these days, human factors is very much part of the equation.

The world's leading aircraft manufacturers are more aware than ever that cockpit layout and pilot comfort are hugely important in giving combatant flyers an edge that may make the difference between success and failure on a mission. The recognition of the importance of human factors is not new, though it is taken into consideration today more than it has ever been. It actually dates back to World War II, when new pilots had to be trained up at speed. Even then, psychologists at Cambridge University led by Kenneth Craik and Professor Sir Frederic Bartlett came to realise that a lot of mistakes aircrew were making were down to the layout of the cockpit.

The dials and controls at that time were fairly rudimentary and they weren't necessarily positioned so that the pilot could quickly and effectively understand and absorb the information being relayed to them through these instruments. A better cockpit layout, it was realised, would lead to a higher success rate. This was an important discovery, and human factors has moved on a long way in the last 80 years. One of the companies leading the way in human factors research is BAE Systems, which operates across the globe in the design, manufacture, upgrade and support of combat and fast jet trainer aircraft including the Hawk and Eurofighter Typhoon.

Suzy Broadbent, Engineering Manager for the Human Factors Shared Service at BAE Systems Air, says that whilst human factors has long been recognised as a valued discipline, understanding of its application for military aircraft and systems has been an evolving story.

"For a long time, military commanders were focused just on picking and training the right people," she explains. "Your skill as a pilot could be based on how good you were at horse riding. Over the years, that changed, and there was a shift in focus to the design of the aircraft. Research began to explore the stress of flying and what was affecting people in that environment. Then in the 1970s and 80s there were a lot of studies to analyse pilot error and what the person at the sharp end was doing wrong."

The Eurofighter Typhoon was the first military aircraft developed in the UK where human factors became a central part of the overall design from the outset. "The cockpit became a subsystem in its own right and had its own dedicated design team. Until then, designers hadn't considered the human factors of the cockpit so much. It was very much driven by functional need. No-one was in charge of the overall layout and making sure it was coherent."

With Typhoon, everything changed. "In the early days of the aircraft's development, mission analysis started, which involved trying to understand what tasks the pilot had to undertake. That meant that the designers knew exactly what information they needed. As aircraft have become more complex, the amount of data available has rapidly increased. It's our job as human factors engineers to ensure the pilot is getting the information they need, at the right time. You shouldn't overload the individual who is having to make all the decisions."

The virtual cockpit moves the information flow from the aircraft directly to the helmet worn by the pilot

Clearly, there is a critical balance to be struck in ensuring that the aircrew are not compromised by having either too much or too little data available to them at a particular moment in time. "It's understanding what the piloting task is and which bits are important at what stage. It's also about controlling the layout of the cockpit and the displays - how they look and how they work - and asking how we can help and what we need to do to ensure that the pilots are less likely to make mistakes and so more likely to succeed in their mission."

So how is this information actually conveyed? In the Eurofighter cockpit, a head-up display projects green imagery such as the horizon line in front of the pilot as a primary flight reference. There are also three different multi-function displays along with the stick and the throttle. "It's quite busy as there are lots of displays and lots of controls. The challenge really comes when the aircraft has to be upgraded to cater for a new weapon or a new sensor. We have to look at ways to incorporate new capability and ensure it's intuitive at the same time."

To address this problem in the future, BAE Systems is developing concepts for a virtual cockpit. Suzy explains, "Everything can be projected onto your visor and as you move your head, different displays will pop up around you. The technologies we have today mean we can have complete flexibility in the design."

The development of the virtual cockpit is ground-breaking, moving the information flow from the aircraft directly to the helmet worn by the pilot, which starts to be a cockpit in its own right. "They would look down and still see a traditional aircraft layout in front of them, but the reality is that it wouldn't be there. The display would all be in the software. That means that when you come to upgrade you have a lot more flexibility and people can set up the information display in the way they prefer. People these days want a lot more customisation and we're used to having information presented to us in different ways - on our mobile phones, for example."

This move to what amounts to wearable instrumentation creates challenges for human factors designers. The

The cockpit of a Royal Air Force Typhoon jet fighter aircraft

We look at which functions we can automate, freeing up the pilot to do the things that humans are good at, like critical decision making

helmet of the future will contain a sophisticated colour display but it must also continue to service its original function as a head protector and life preserver. "It still has to be able to support pilots if they have to eject which means we have to take into account the comfort and the weight. We're also examining the possibility of building an eye tracker into it so that we can see exactly what the pilot is looking at. That would be useful for training; if we can see that they're looking at the right things when they land, that will help with the debrief. We'll also be able to gather information about how tired they were."

Suzy continues: "There's a lot you can learn from the eye and this technology means we'll be able to tell more about a mission than simply listening to the pilot's subjective view. We'll know if they were looking at the right thing or if they've been fatigued or overloaded. Might they potentially be pre-hypoxic? Are they suffering from g-forces and the risk of a loss of consciousness? This information can be fed back and evaluated in real time. We've long had warning panels telling us which bits of the aircraft may be having problems but we've never had the same thing for the pilot."

Suzy describes this as human-machine teaming. "You don't want the pilot to be the weak link. You want the technology to be able to step in and help them where necessary, especially as modern aircraft only have one person in the cockpit. What we're doing is looking at which functions we can automate, so freeing up the pilot to do the things that humans are good at, such as critical decision making and the ethics of a particular situation. The machine can take over a lot more of the flying, especially if the pilot does start to get overloaded."

Using this new technology, the mission becomes more than the product of a single machine and a single human. Of course, that's always been the case - the Battle of Britain was won by commanders and squadrons rather than lone Spitfires and Hurricanes operating independently - but the technology will allow more agents to get involved. One possibility is the future use of so-called loyal wingmen - unmanned devices such as drones flying alongside the manned aircraft that can be sent off to do other things.

"If the helmet fails and that's where all the displays are, then clearly there has to be some redundancy. In a worst case scenario, someone on the ground could take over your aircraft, or it could even fly you home itself."

Inevitably, given the complexity of modern military aircraft and the critical importance of getting the design of everything right, systems and technology take time to develop. The work being done now will probably not actually be seen in cockpits for at least another decade and possibly longer.

"We don't know what the missions will look like in the future or exactly what the aircraft are going to be like then but we can start to build in flexibility now."

This may all sound like science fiction, but so did autonomous cars and drone deliveries a decade ago, and while they're still in development, both are a reality now. It's not likely to be long before the intelligent helmet and the associated technology move out of the design and test phase into real world combat air capability.

"There are always technological challenges, and there are always hundreds of reasons you can give why something can't be done," Suzy says. "We also have to take the customers and regulators on this journey with us but it's all very exciting and we've got some great people on the teams involved. At the end of the day, this is all about using human factors understanding to help the pilots. They shouldn't have to fight against the systems to get the information they need. And the whole experience should be intuitive and engaging for them. We still want people to want the job."

Suzy Broadbent is Engineering Manager for the Human Factors Shared Service at BAE Systems Air.

Responding to a crisis

The need for civic authorities to meet the challenges of local disaster planning has risen, as highlighted by the current pandemic, recent terrorist attacks and natural disasters. With their experience in the aftermath of the Grenfell fire, **Courtney Grant** and **Mohammed Ali** describe how a human factors perspective can help

On 14 June 2017, 72 people lost their lives in an catastrophic fire. In the days after the disaster, we acted as volunteers for the National Zakat Foundation (NZF), a charity that formed part of the Grenfell Muslim Response Unit. We provided immediate help and support to the survivors and other affected local residents, and the experience allowed us to reflect on areas where human factors could play a key role in helping civic authorities respond to a crisis of this nature, and to mitigate the challenges faced.

The NZF is a Muslim faith-based charity, that primarily collects 'Zakat' and uses it to support local welfare initiatives. Zakat is an annual payment of 2.5% of a Muslim's accumulated total wealth over a one-year period that all eligible Muslims make under Islamic Law. In order to offer recipients dignity, independence and choice, NZF handed out cash sums to individuals affected by the Grenfell tragedy.

The NZF used crowdtasking to recruit volunteers, leveraging NZF's social capital influence within the third sector and the Muslim community. As a volunteer based at Al Manar mosque, located a mile from the Tower, Mohammed processed financial aid to survivors. Volunteers used spreadsheets to manage different funds available to survivors. (They also carried out health needs assessments via a smart phone app.)

The system used three pools of funds: the charity fund, the 'Zakat' fund and the *Evening Standard's* dispossessed fund that created a complex situation in which to manage the money. The Zakat fund had to be exclusively distributed to Muslims so volunteers had to determine a recipient's faith, which wasn't always easy. NZF could not distribute any surplus funds that the Al Manar mosque collected. And the *Evening Standard* allocated their dispossessed funds to two charities: NZF and Rugby Portobello Trust. Grenfell Tower residents had exclusive rights to this pool of funds but could only claim once from either of the two charities. Whilst a live spreadsheet was used to document and authenticate the status of each Grenfell resident's payment, it didn't always have the most up-to-date information. Having a simple one-pool system that provided immediate feedback on the status of claims would have eliminated any confusion and speeded up the process.

The cognitive tasks of volunteers were often complex too. Given the nature of the tragedy, some victims had no identity documents and volunteers had to make decisions about payments based on judgement and intuition rather than on verification. Volunteers also provided counselling services to victims whether or not they had the necessary skills, highlighting an important training gap. Journalists entered the mosque in an attempt to speak directly with survivors so volunteers had to divide their attention between helping survivors and tracking the movements of the journalists.

NZF had a help desk at the Westway sports centre located close to the Tower. Courtney provided survivors and other affected local residents with advisory support but simple solutions could have made this easier. For example, one family had very understandably left their mobile phone charger behind when escaping from the Tower. It took a lot of time to find the right charger for them to use but having a centralised system that allowed volunteers to quickly search for key items like this would have helped enormously.

Some people resided at the Westway, whilst others stayed elsewhere but visited the centre to receive support. Analysing these different situations would have helped to understand the various needs. Mapping the end-to-end process in a systematic way and looking at the patterns that emerged could have better informed the layout and arrangement of support desks to ensure those in need found what they wanted quickly and easily.

In the face of the disaster, the charities contributed in making a positive social impact and filled a void in the absence of wider coordinated support. Local authority initiatives have since been implemented which have been inspired by the work of such charities. This led to the establishment of the Coordinating And Mobilising Emergency Response Activists (CAMERA) across London boroughs. In such trying circumstances, it's no surprise that issues emerged and it shone a light on areas where human factors could help to respond to a crisis of this nature in the future.

Grenfell Tower in the aftermath of the fire

IMAGE: SHUTTERSTOCK

Courtney Grant has a BA (Hons) in Psychology, an MSc in HCI with Ergonomics and is a Chartered Fellow of the CIEHF and a registered European Ergonomist.

Mohammed Ali has a BA (Hons) in Accountancy and Business and an MSc in Global Health Policy. He's a member of the CAMERA Emergency Volunteer Team in Hammersmith & Fulham.

Finding a fix for maintenance errors

Despite many years of regulation, focus and effort, aircraft maintenance still experiences incidents and accidents with common contributory factors. **Jo White** and **Sarah Flaherty** ask whether we're learning sufficient lessons regarding performance influencing factors

Whilst the number of fatal accidents attributable to maintenance has declined, a review of data shows that the industry still faces issues with procedures not being followed and components being installed incorrectly. An in-depth investigation of any significant incident takes time and resource but without a systematic investigation, organisations can often make assumptions about why people behaved in the way that they did during an event.

A candid discussion of a real-life example provided by Charlie Brown, Safety and Compliance Manager at Virgin Atlantic Airways, related to an incident that proved a timely reminder to the organisation of the fallibilities of systems and people.

An aircraft had departed from Gatwick Airport for a scheduled flight to Las Vegas. Following retraction of the landing gear after take-off, low quantity and pressure warnings occurred on a hydraulic system, due to a hydraulic fluid leak. The required checklists were completed and the aircraft returned to land at Gatwick. As the landing gear extended during the approach, the right wing landing gear struck the gear door, preventing the gear leg from fully deploying. The crew carried out a go-around and, following a period of troubleshooting and associated preparation, a partial gear landing was successfully completed. It was subsequently determined that the hydraulic retract actuator on the right wing landing gear had been incorrectly installed but had gone undetected, preventing the leg from fully deploying.

An investigation identified a number of contributory factors relating to the incident including that the procedure was not written in accordance with human factors best practice, and the language

used was confusing. Following the occurrence, a number of recommendations were made including that all future retract actuator replacements require the addition of a fluorescent tape strip affixed to the down facing side of the actuator, clearly annotated in black permanent marker with the words 'THIS SIDE DOWN' once the correct orientation of the actuator is established. This was to be carried out during the 'bench' preparation stage when all the hydraulic hardware is installed and checked, and is an ongoing requirement.

In addition, there was an immediate but temporary requirement for full gear retraction functional tests (swings) to be carried out after the hydraulic retract actuator replacement to prevent a further incident. The delay gaining access to sub-assembly stores, a lack of information explaining how to best utilise manual handling aids such as a sling and hoist which resulted in aids not being considered, a lack of readily available correct tooling, the handover of the task to the night shift, manpower concerns, and the failure of the Strike Board to position correctly, all in turn, lead to the partial gear landing.

The investigation further concluded that the task became so physically demanding that the maintenance team had become entirely focused on just attaching the actuator to the aircraft, in order to relieve themselves of the 85kg weight they'd had to manually support for over 30 minutes. As such, they had no remaining capacity to ensure they installed the actuator in the correct orientation. It was subsequently determined that they had rotated it 180° about its long axis during installation, effectively installing it upside down. A difficult lesson learnt which was recommended to be shared.

The Royal Aeronautical Society (RAeS) Human Factors Group: Engineering (HFG:E) is an active subgroup of the RAeS Human Factors Specialist Group, with the purpose of promoting and influencing the reduction of risks to airworthiness resulting from human performance in engineering. Last year, through a conference at Cranfield University, HFG:E posed the question to industry about whether or not we're learning from maintenance error.

At the event, it was suggested that commercial pressures can impact the depth of human factors investigations as well as any subsequent investment in corrective actions and interventions. CAA Airworthiness Surveyor Dr Marie Langer described an analysis of Mandatory Occurrence Reports and CAA Airworthiness findings over the past decade that suggested trends remain largely unchanged in terms of the number and types of findings and occurrences. She noted that the industry understanding of human performance, the quality of investigations and proposed corrective actions varies considerably, particularly the ability to identify underlying issues.

Dr Langer said that investigations often stop after the first visible cause is identified, rather than seeking the root cause. With 44% of all CAA findings raised in the previous five years relating to recurring incidents, a CAA root cause analysis identification road map was proposed to assist organisations in establishing a suitable process for investigation. Investment in the tool should lead to operational efficiencies in the long term, provided it is used effectively, reducing the recurrence of incidents. Fundamental to the CAA's suggested process was monitoring the outcome of any preventative actions to assess whether the intervention has been effective.

IMAGE: GETTY

An aircraft engineer repairing and maintaining a jet engine

Andy Evans, Director of Aerossurance, said that just sending people through more human factors training is not always the answer to improving maintenance. He suggested instead, "we have to share the knowledge currently locked away in safety reporting databases more widely and engage our workforce more proactively to create improvement rather than just passively waiting for safety reports". More recent Military Aviation Authority regulated human factors facilitation courses also provide a forum to share some of this valuable information.

Evans advocated a more enlightened and holistic 'Psychological Safety' approach that encourages open discussion and confidence in the reporting system and management

Humans are normally not 'hazards' to be managed and disciplined but are 'heroes' who can help our organisations learn

approach to investigating and understanding events. Humans are normally not 'hazards' to be managed and disciplined but are 'heroes' who can help our organisations learn and so we should treat them accordingly using 'Just Culture'.

The HeliOffshore Human Hazard Analysis process, that gets designers talking with maintenance personnel, and the Flight Safety Foundation Maintenance Observation Programme, that looks for improvement opportunities with maintainers before errors occur, are both examples of proactive safety management.

The conference speakers highlighted that as an industry, we are learning but perhaps not at a speed nor to a depth and consistency to create a step change in the safety performance of maintenance organisations. There is clearly more to be done.

Joanna White is a human performance specialist with experience across army and aviation projects and has spent her career working with both civil and military organisations. She has a first degree in psychology and an MSc in Research Methods in psychology. She is a Chartered member of the CIEHF and sits on the Royal Aeronautical Society Human Factors Working Group: Engineering.

Dr Sarah Flaherty is Director of Lux Consulting and is a human performance specialist with a broad experience across the aviation, rail and petrochemical industries. She has a first degree in psychology and is a Chartered member of the British Psychological Society and the CIEHF.

Further reading

CAP1760, Effective Problem Solving and Root Cause Identification: https://bit.ly/38WVVG5

The wrestler, the bus driver and the soldier

In the developed world, health and safety interventions have been very effective at reducing workplace deaths. In the UK, the rate of fatal workplace injuries has reduced by 85% between 1974 and 2019. But, as **Pat Jordan** discusses, workplace injuries are not the only way that work can kill us

A friend who was a professional wrestler invited me to a show. That evening, along with 18,000 others, I watched muscular men and athletic women throwing each other around the ring. My friend won her match after hitting her opponent over the head with a chair! It seemed violent and dangerous at the time but professional wrestlers are highly trained 'sports entertainers', working together to ensure that, while everything looks spectacular, they take care of each other's safety. Nevertheless, an alarming number die young.

WrestleMania is wrestling's annual flagship show. Thirty-nine wrestlers took part in the 1990 event. By 2015, fifteen had died. To put this in perspective, of the 44 American footballers that took part in that year's Superbowl only one had passed by 2015. That gives these wrestlers a death rate of 17 times that of the footballers. But no WrestleMania 1990 participant died while wrestling, training for wrestling or of a wrestling injury.

Studies have shown that the Standardised Mortality Rate (SMR) of pro wrestlers is much higher than that of the population as a whole. In other words, a wrestler is far more likely to die in any given year than the average person of the same age and gender.

STANDARDISED MORTALITY RATE

Standardised Mortality Rate (SMR) is a measure of deaths per 100,000 person-years adjusted for age and gender. It provides a meaningful comparison of the death rates of different professions, even if the age and gender composition of those professions differs.

Having made such progress in reducing workplace deaths, perhaps the next stage is to engage with a wider measure of safety and embrace broader approaches to prevent people being killed by their jobs.

Current health and safety analyses define a work-related death as one occurring from injuries sustained in the course of work. On this basis professional wrestling might not be regarded as an especially dangerous job. But while they were not killed at work, lifestyle factors surrounding the job might explain these deaths. These include using steroids and painkillers, and hard-partying associated with touring.

When assessing the dangers of work, SMR can be a complimentary measure to workplace fatalities. The latter are at an all-time low but are apparently safe jobs killing people in other ways? For a government study, I investigated occupational hazards and workers' health. A memorable visit was to a bus company. No bus drivers had been killed or injured at work but they had a noticeable problem with obesity.

Bus driving is very sedentary, without even the movement that an office worker might get, for example when they get up to make tea or chat with colleagues. One employee remarked, "A number of mechanics have moved into driving and you just see them expand."

Most recent figures for male bus drivers show about two workplace fatalities per 100,000 person years, as compared to one for the male workforce as a whole. The SMR of male bus drivers, meanwhile, is 445 per 100,000 person-years compared with 384 for the working population. Taking these figures together suggests 61 additional deaths, only one of which is caused by a workplace fatality.

This SMR figure comes from a paper in the *Lancet*, which

analysed mortality rates for 63 separate occupations. Occupation-based SMR figures used to be published by the UK government every decade, but this stopped in 1970, since when work has changed enormously. Figures have been produced in other countries but usually based on broad occupational categories rather than specific jobs, limiting their usefulness.

The only organisation that uses SMR as their main measure of hazard is the Ministry of Defence. Their data shows that even during the Iraq and Afghanistan wars, military personnel had low mortality rates. Currently these are 40% of that of the UK population as a whole. Traditional health and safety analyses would suggest that being in the military was by far the most dangerous job in the UK. Between 2001 and 2014, 689 UK military personnel were killed in conflict and there were also fatalities during training. Even now the military has by far the highest number of deaths caused by accidents at work, at around 14 per 100,000 person years, way ahead of agriculture in second, which has eight.

Having made progress in reducing workplace deaths, perhaps the next stage is to engage with a wider measure of safety

Given these figures, why do military personnel have such low mortality rates? The answer may lie in lifestyle. Military personnel lead active lives, stay fit and have regular medical check-ups. Camaraderie reduces loneliness and mitigates other issues effecting longevity. The military has also got better at addressing conditions such as Post Traumatic Stress Disorder. Having said that, people need to be fit and healthy to join the military in the first place.

SELECTED OCCUPATIONS BY MALE SMR

Armed Forces personnel	162
Doctors	225
Teachers	262
Engineering professionals	282
Electricians	333
Farmers	402
Sales assistants	429
Bus drivers	445
Hairdressers	492
Professional wrestlers	708

SMR (DEATHS PER 100,000 PERSON-YEARS)
NOTE: *THE LANCET* DATA COVERS 63 OCCUPATIONS BY MALE SMR, 43 FEMALE.

This last point illustrates the limitations of SMR as a measure. It helps flag jobs where people have high death rates but doesn't show causes or even whether there is a causal effect at all. For bus drivers we might expect that the sedentary nature of the job leads to increased mortality but further investigation would be required to know what proportion of additional deaths are associated with it.

For example, male hairdressers have high mortality rates, however this may mainly be due to high numbers of gay men in the profession rather than anything to do with the job. Research by LGBT organisations has shown that gay men have shorter life expectancies and, coincidentally, are overrepresented in hairdressing.

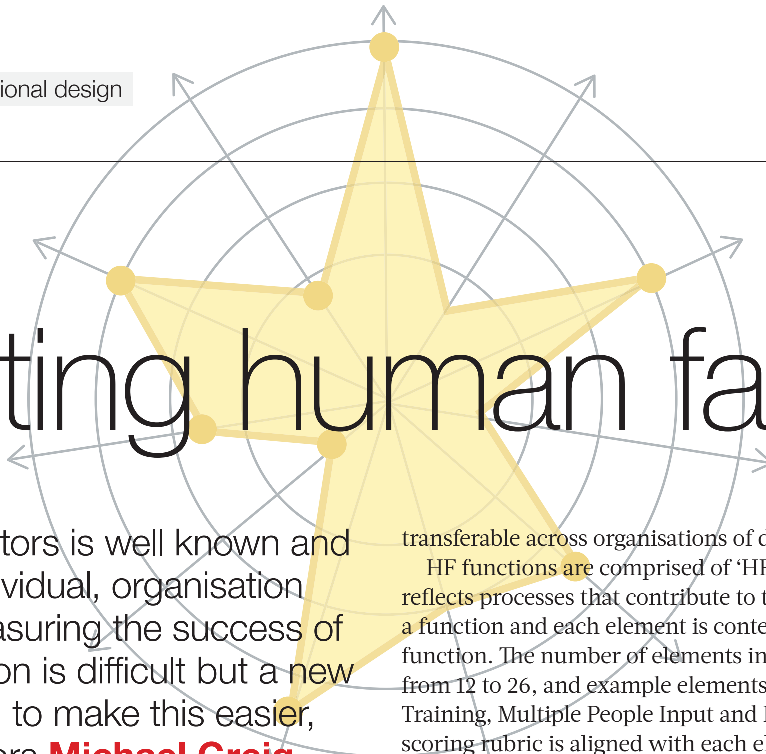
For all professions, there will likely be other correlating factors, such as socio-economic status, which SMR doesn't control for. Whether this should be seen as job-related is debatable. Arguably, a person's socio-economic status is conferred by their job but it's also a predictor of job opportunity.

Despite its limitations, SMR is a valuable measure. Accidents and injuries at work explain a tiny fraction of deaths associated with particular jobs and SMR can show us where to look to help us understand the others. As ergonomists it's our responsibility to press for the reintroduction of the gathering and publication of this data by government. In the meantime, the *Lancet* article, while not covering all occupations, is a valuable resource.

Pat Jordan is an ergonomist working in the private sector, government and academia.

Further reading

Katikireddi, S V et al, 2017. Patterns of mortality by occupations in the UK, 1991-2011: a comparative analysis of linked census and mortality records, *Lancet Public Health* 2017; 2:e501–12



Integrating human factors

The value of human factors is well known and benefits exist at the individual, organisation and societal levels. Measuring the success of human factors integration is difficult but a new tool has been designed to make this easier, as award-winning authors **Michael Greig** and **Patrick Neumann** explain

The application of human factors in organisations is not always ideal and often limited to health and safety domains and retrofitting of equipment or workstations; rarely is human factors applied at the design level. Audits and standards provide some guidance on how to measure the effectiveness of human factors but nothing has been available to know how well it's been integrated across the whole organisation.

The Human Factors Integration Toolset (HFIT) has been created to address this issue, enabling a macroergonomics approach to quantifying the level of integration of human factors throughout an organisation.

The measurement of the extent to which human factors has been integrated and applied in an organisation is based on the principle that decision-makers manage using quantitative indicators and therefore need an approach that allows measurement of human factors integration. Outcomes from this novel approach provide tangible information which can create avenues for discussion on human factors and the associated benefits.

The HFIT was developed with contributions and feedback from an electronics industry partner. Public workshops were held to provide a venue for the content, format and usability of the tool to be verified by subject matter experts. The workshops included participants from the human factors profession, industrial engineering, as well as graduate students, with backgrounds in consulting, healthcare and manufacturing, among others.

To facilitate the review of an organisation, the tool is structured so that an organisation is broken down into 16 'HF functions' to make analysis of an organisation more manageable. Examples of functions are Environmental Health & Safety, Tooling, Finance and Quality. Dividing an organisation in this manner allows for a description of roles in an organisation as compared to using explicitly defined departments since a small company might not have a 'shipping department' for example. This choice was to allow the approach to be transferable across organisations of differing size and structure.

HF functions are comprised of 'HF elements'. An HF element reflects processes that contribute to the ideal human factors for a function and each element is contextualised to the associated function. The number of elements in a given function ranges from 12 to 26, and example elements include HF Specific Training, Multiple People Input and HF Culture. A five-level scoring rubric is aligned with each element and is scaled to reflect the integration maturity of the element from nothing to the perceived human factors ideal of 'world class'.

The HFIT can be completed individually or in groups using self-reflection, interviews or a consensus approach to the scoring. The tool is modular, meaning a single function can be assessed in isolation. Users review each element, choose the appropriate rubric score and then sum the element scores for the function to determine the percentage of the human factors ideal in the current situation. A radar plot shows how the scores of the functions relate in all functions across the organisation.

The tool outcomes can provide an external benchmark for comparison with other organisations which will hopefully spur on 'friendly' competition between organisations and drive the overall work experience to a better state. The scores can also be used internally to track continuous improvement goals over time. Consultants could use the approach and apply the information to demonstrate and/or measure project outcomes and improvement at the organisational level and the tool can provide a platform for discussions around the contributions of human factors in the broader scope of an organisation.

The tool is ready for broad testing to understand how well the approach transfers across different organisations and how well the scores relate to organisational success. For more information about this innovative approach to measuring the quality of your organisation's human factors capabilities, please contact **m2greig@ryerson.ca.**


Michael Greig and **W Patrick Neumann** work in the Human Factors Engineering Lab of Ryerson University in Toronto, Canada.

THE AWARD-WINNING PAPER!

The description of the development of the HFIT has recently been awarded the 2020 Liberty Mutual Award for best paper in the journal *Ergonomics*, and is available at

www.tandfonline.com/doi/full/10.1080/00140139.2019.1572228.

The tool is available for free at **www.researchgate.net/project/Human-Factors-Integration-Toolset.**

Getting engaged

We all have different personalities and some of us are more courageous than others, especially when it comes to sharing knowledge or asking questions. During our latest annual conference and numerous other virtual events since then, we've seen a huge demand for exchanging opinions, making connections or simply saying "nice to see you around". Some members reconnected after having worked together in the past, others are actively looking for new links. Whatever your motivation may be, our organisation provides lots of opportunities to get engaged so if you haven't done so already, please explore what's on offer.

Whether you're getting started in a new discipline, seeking collaborators for a new project, you'd like to learn from other sectors or hear more about human factors in other countries, you'll benefit from networking. It may help you to complete the paper you're working on, find a new project or gain another perspective but it can also contribute to making your career more resilient and open up new opportunities.

Our discussion forum Communities is available to all members and has seen a huge increase in posts over the last few weeks. Time zones are not an issue as discussions can easily ping-pong across the globe, so please join in wherever you're located. The forum not only addresses immediate questions but also creates a reference resource which all members can benefit from at any time.

Other engagement opportunities for members are being created by our Regional Network Leads who are planning virtual and (as soon as possible again) in-person events, so you can make contact with other members in your area. If you'd like to get in touch with your Regional Network Lead go to www.ergonomics.org.uk > Get Involved > Regional Networks.

Our online events programme is building up and all events have a chat function so if you can attend you can get involved this way, by interacting with the presenters or other participants.

Find out more on our website or contact me for any further information.

Iris Mynott
i.mynott@ergonomics.org.uk
07702 542166

MEMBERSHIP

Our latest accreditations

Congratulations to the following members whose applications for accreditation by the CIEHF over the past few months have been successful. Registered Members and Fellows also have Chartered status.

Fellowship
- Laura Lewis
- Somnath Gangopadhyay

Registered Membership
- Paul Isherwood
- Kate Shield
- Isabel Burrows
- Linton Seabrook
- Evanthia Giagloglou
- Ane Merino Guevara
- Harry Conway
- Katie Plant
- Pak Ho Plato Chan
- Jonathan Mason
- Jody Burley
- James Bunn
- Anisha Tailor

Technical Membership
- Rachel Bennett
- Mark Benton
- Andrew Wright

CIEHF events at a glance

Eligible for CPD Lifelong Learning

For more details of all CIEHF events, see our website at **events.ergonomics.org.uk**

EVENT	WHEN & WHERE	DETAILS
Postural Analysis Tools Masterclass	Wed, 26 August, online	Learn, explore and practice use of human factors tools to measure and analyse posture. First in a new series of expert-led masterclasses.
Aviation Safety	Wed, 23 September, online	Launch and discussion about CIEHF's upcoming White Paper containing visions of 15 thought leaders, showing how they believe aviation evolution will unfold between now and 2050.
STAMP Masterclass	Fri, 25 September, online	Find out about this human factors tool that looks into causes of incidents by analysing complex processes and unsafe interactions among system components.
IEA2021	Sun-Fri, 13-18 June 2021, Vancouver	Learning and networking opportunities for the human factors and ergonomics community around the world. Call for submissions closes 25 September. Visit www.iea2021.org for details.

Please note that some events details may be subject to change after publication. Please check the events website for up-to-date information.

IMAGE: SHUTTERSTOCK

Q&A MEMBER PROFILE

Tina Worthy talks to Ergonomics Advisor **Siobhan Burns** about her career in healthcare

IMAGES: SHUTTERSTOCK / ALAMY

What's your background?

Like many who find a career in ergonomics and human factors, I didn't follow a planned route. I started out as a Healthcare Assistant working in a nursing home specialising in patients requiring palliative care. No patient handling training was given to staff at this time and within two years I had a significant back injury due to the weight and number of patients that I moved every shift.

Through that, I developed an interest in safe patient handling and completed local training to become a manual handling trainer specialising in patient handling. I worked in this role in a variety of settings for just over six years until I became a Manual Handling Trainer at University Hospitals Birmingham NHS foundations Trust (UHB) in summer 2013. This was the turning point for me, as I suddenly had a manager who was very proactive at developing her staff. Within no time I was booked onto a Postgraduate Diploma in Ergonomics for Health and Community Care and was feeling nervous about being back in an academic setting after so many years.

But I found the experience enlightening and interesting, and I particularly enjoyed the Human Factors and Systems module. I was also very lucky to work alongside, for a day or two each month, an inspirational colleague, Frances Ives, who was the Ergonomics Adviser for UHB at that point. Her down to earth humour and obvious dedication to integrating good ergonomics into everything she touched (including her Christmas gift list), helped me through the difficult times and motivated me to study harder.

What made you apply for your new role as Ergonomics Adviser?

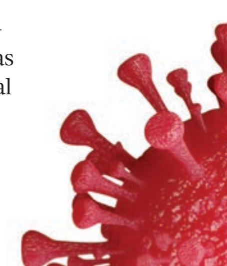
Working alongside Fran enabled me to see some of the elements of her job and gain a deeper understanding of how an Ergonomics Adviser can help to improve the working lives and environments of staff. It ignited a desire to continue on my ergonomics journey and, whilst I continued to have a passion for manual handling, I was keen for a new challenge.

Fran was also moving on and, as this coincided with the conclusion of my postgraduate qualification, she encouraged me to apply for the job. The thought of taking on such a challenging role and following in the footsteps of someone I admired was certainly daunting but such an opportunity rarely comes along and I felt that I had nothing to lose in applying.

How did the outbreak of Covid-19 affect your work?

My first day in the Ergonomics Adviser role was 1st March 2020, and the ominous rumblings of Covid-19 were approaching. Three weeks in and lockdown was announced which completely changed the landscape of my job. Virtually all requests for ergonomics input dried up and I was forced to cancel all my booked appointments, as staff were either sent home to shield or re-purposed into different roles.

The atmosphere at work was tense and nerve-wracking as nobody knew what the next day was going to bring. I was pulled back into the manual handling team and assisted with the training of



over 1000 staff in preparation for the possible opening of the Birmingham Nightingale Hospital. For about ten weeks I worked in a weird limbo of mostly manual handling, with occasional dips into ergonomics work.

Finally, in June I was able to start picking my ergonomics work back up and have been venturing out into a changed environment. Assessments that used to be a relaxed chat have morphed into a masked and gloved odyssey, with the participants nervous about sharing airspace, and disinfectant wipes our constant companions. Many face-to-face meetings are now performed via webcam or phone which brings with it new difficulties when connections are unreliable and private workspaces are at a premium. Working in this way, although undeniably necessary in the current climate, can reduce the quality of information I get from the person being assessed.

What projects have you been involved in so far?

Whilst delivering patient handling training at the Nightingale Hospital, I had an insight into the behind-the-scenes work involved in putting the facility together. I was able to offer support and guidance to staff who were feeling overwhelmed and tired. Due to the speed at which everything moved, very little consideration had been given to the comfort of staff working with display screen equipment. Most were sitting on hard, non-adjustable seating at tables that threatened to give them splinters every time they moved their hands. It gave me an enormous sense of satisfaction when I was able to source some basic operator's chairs and give them to the worst affected workers. I was also involved in ensuring that the battered tables were replaced with proper desks.

I've also been heavily involved in the management of homeworkers. For UHB, homeworking wasn't the norm prior to Covid-19, so as a Trust we weren't prepared for the mass migration of staff home. I've been involved in ensuring that the Trust is meeting its legal duties for staff and also supporting workers who are finding homeworking difficult.

What challenges do you see in the future?

One of the challenges is the management of homeworkers when they have workstation concerns. Pre-Covid I would have face-to-face meetings with staff who were struggling and see their workstations, observe postures and how they interacted with their environments. This made it much easier to identify problems and suggest changes, whereas now, such visits are not a viable option in many cases. While we use applications such as Vidyo and Microsoft Teams, it's not the same as being there and makes it harder to observe 'work as done'.

With so much money having been poured into the management of the Covid-19 situation, it's almost certain that there will be a reduction in funding for human factors work. For the uninitiated, incorporating good ergonomics into a project can seem like a luxury, especially if easier, cheaper options are readily available. I'll need to be convincing in my arguments around the value of the ergonomics service.

Have you got any advice for others in healthcare who'd like to follow the same path?

Find yourself an approachable, experienced mentor. Having someone that you can turn to when you have questions, who's happy to answer the most basic queries is so helpful. Sometimes tasks are very daunting and it's hard to know where to start. Having someone who's been there before is absolutely invaluable.

Compared to other sectors such as nuclear and aviation, healthcare lags behind in its application of ergonomics to working procedures. This often means that people don't see the value in the application of good ergonomics, but don't give up! The pleasure in seeing people working more comfortably and in safer environments is worth all the hard work.

Three weeks in and lockdown completely changed the landscape of my job

Success stories

In our second set of stories from this year's amazing Institute award winners, we celebrate the achievements of members for their work in using virtual reality to prepare for real-life scenarios, for their creative approach to communicating reports and for generously volunteering their time to maintain professional standards.

JOHN LOVEGROVE, FOUNDER OF CANARY DESIGNS

Outstanding Communications Award

John Lovegrove's ergonomic design studio is one of only a handful in the UK to use 3D environment software together with digital manikins which can model human interactions within a CAD design framework. This enables John and his team to immerse themselves in the design and test out all the human factors criteria in a physical space simulation and to quickly propose ergonomic enhancements.

This ability to work quickly and visually has shaped his communications with clients. Instead of providing large hefty reports, John creates video summaries of written reports, which allow his clients to understand the key findings quickly and effectively. Canary Designs' website also hosts a growing selection of video tutorials which explain human factors concepts and the benefits of virtual simulation and using digital avatars. As a result of this innovative approach, Canary Designs has been recognised with CIEHF's Outstanding Communications Award for 2020.

John has taken this approach over several years following client feedback that they were suffering from information overload. He said: "People have not got the time to read big in-depth reports, so we break down the elements of our report so clients can pick out what they need, and creating video summaries is a good way of getting your message across as quickly and concisely as possible. And secondly, it's a good way of showcasing the kind of work we are doing to the rest of the HF community, particularly with the computer modelling techniques. Once you have the CAD model for a project you can look at the design from a human factors perspective before you meet anyone from the project or visit the site. This means I can go to the first client meeting with a whole set of pertinent questions and potential solutions, saving everyone time.

John said the modelling technique is also a powerful way to put human factors arguments across at meetings, as he explained: "This approach allows you to put your ideas across in the universal language of design to multiple stakeholders. I've used it in a room full of officers for a military vehicle project showing the ergonomic adjustments required to create a more effective emergency escape route for occupants. It required significant engineering changes but it's the first time I've ever had full agreement across the entire room. That's the beauty of this visual method."

John and his team are working on other communication projects to enhance the

Creating video summaries is a good way of getting your message across as quickly and concisely as possible

Using computer-aided design to demonstrate workplace layout and work processes

business: a linked website that provides online ergonomic self-assessments for organisations with tutorial videos, and eight new videos to enhance his current website to explain his services in more detail.

Commenting of the award, John said: "It's great to win the award, and to be recognised by my own community as communications is so important to all of us. I hope other human factors professionals will look at using the types of techniques we use at Canary Designs.

IMAGES: ISTOCK / GETTY

MIKE GRAY

Volunteer of the Year Award

A mine rescue team gets to work

The young Oxford physics graduate, Mike Gray, was first introduced to ergonomics in the 1970s when designing breathing equipment for mine rescue teams, and his interest in this new discipline helped launch his career in ergonomics which has now spanned over 40 years. During his time working at the Health and Safety Executive (HSE), he has also been heavily involved in the Institute, working on its various panels and boards. This long service has been recognised and celebrated by Mike being named as the recipient of the CIEHF's Volunteer of the Year Award for 2020.

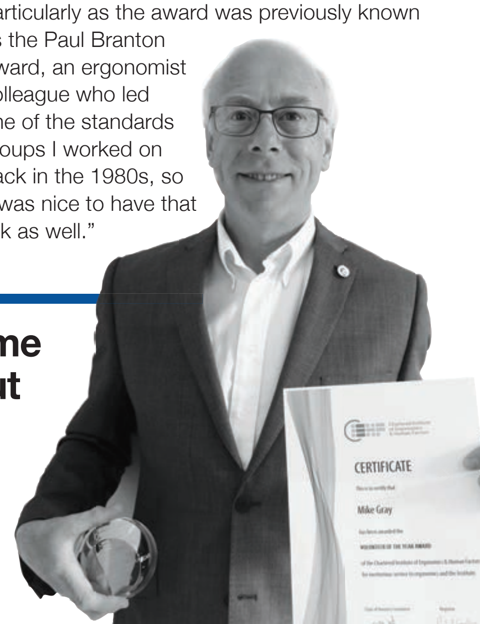
Mike said: "It was when we were trialling the new coal mine rescue equipment underground, and looking at what heat tolerance people could take, that I became interested in human factors. The HSE could see that they needed to take account of this discipline so they supported me to study a Masters in ergonomics, and that became my new profession."

Mike helped to establish a new team of ergonomists at the HSE and later worked as an ergonomics specialist inspector, building up a team of frontline specialists across the country. He has also been involved in accident investigation, providing technical guidance to industry and helping to establish international standards.

He first joined CIEHF in the early 1980s (when it was known as the Ergonomics Society) and got involved in the Institute's health and safety Special Interest Group set up to provide guidance on various government proposals and standards. He was also involved in organising the Institute's 50th anniversary event at the London Science Museum and has subsequently served on the CIEHF Council, Professional Affairs Board (PAB) and as the UK representative on the Council for the Registration of European Ergonomists (CREE).

Although Mike retired four years ago, he still volunteers on PAB and CREE. He said: "I give my time as it helps to keep me involved in my profession but to also allow me to pass on some of my experience. I think that the Institute could make more use of retired members as we have a little bit more time on our hands and great experience, which I think is a valuable resource to capitalise on."

Commenting on his award, Mike said the announcement came out of the blue: "I really had no idea and so I feel very privileged to win, particularly as the award was previously known as the Paul Branton Award, an ergonomist colleague who led one of the standards groups I worked on back in the 1980s, so it was nice to have that link as well."

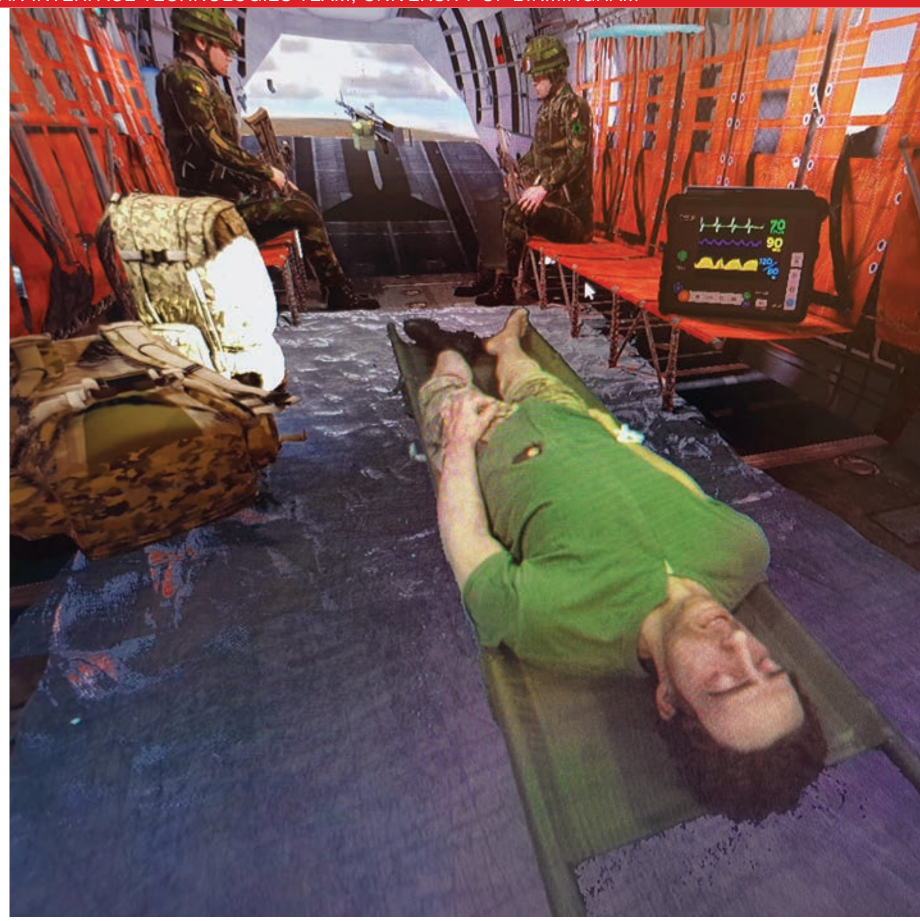


Volunteering helps to keep me involved in my profession but also allows me to pass on some of my experience

PROFESSOR BOB STONE, DIRECTOR OF THE HUMAN INTERFACE TECHNOLOGIES TEAM, UNIVERSITY OF BIRMINGHAM

The Innovation Award

The quality of the emergency treatment that can be given to a critically wounded soldier from the extraction point in the conflict zone en route to a hospital base can be the difference between life and death. That's why Medical Emergency Response Teams (MERTs) are put through a range of pre-deployment training scenarios to condition them to the difficult conditions they could encounter working in the field.



Example of real and virtual integration in a MR Simulator

In the past, this relatively expensive training has been carried out on static structures in military bases across the UK but now, thanks to Professor Bob Stone and his Human Interface Technologies Team at the University of Birmingham, pre-deployment training has the potential to become more cost effective and more 'real'. That's why they have been awarded CIEHF's Innovation Award for their Mixed Reality Project for the Medical Directorate of the UK's Joint Medical Command.

The team has developed a low-cost redeployable training concept for MERTs which uses 'mixed reality'. This involves using real physical objects – such as realistic human manikin – within a 'blue screen' virtual reality (VR) enclosure. By using VR headsets, medics can believe they are providing life-saving medical procedures in a number of military transport scenarios, such as the back of a Chinook helicopter, on the floor of an armoured personal carrier, on the deck of landing craft or even inside a hovercraft.

The mixed reality technology allows them to interact with a virtual patient, but at the same time see and touch the physical manikin, while being surrounded by the sights, sounds and smells of that particular vehicle, including moving avatars of other crew members.

The instructor is in constant contact with the trainees, observing their medical procedures but also recording their reactions when he decides to throw in some gunfire, an engine failure or even a dust storm. The VR headsets also track eye movements, showing what the trainees are focused on and when they are being distracted.

Professor Stone's team is currently developing an augmented reality tool designed to portray the sudden appearance of wounds on the manikin, such as bleeding or burn blisters, so that medical crews will have to react immediately.

Our concept is a cost effective way to expose medical personnel to incidences they could encounter in the theatre of war

Professor Stone said: "Evidence from Afghanistan and Iraq shows the importance of preparing medical personnel for the events they could encounter in the theatre of war, so our concept is a cost effective way to expose them to some of the incidences they could encounter in a realistic setting. Throughout the whole project we have taken a human factors approach, and at every stage involved the trainers and medics to ensure the concept is right for their needs. It obviously has many civilian applications too but the next stage is to carry out intensive trials at military bases around the UK to find out what medics learn under these conditions and, more importantly, what they retain and transfer into a real world setting."

Commenting on the award, Professor Stone added: "I'm fantastically pleased to get the award and so proud of my team. We want to prove that by adopting a human factors approach at the outset that you can deliver a technology-based training system that's fit for purpose."







Human Science Research

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K Sharp Ltd - Head Office
Suite 2, West End Yard
21-25 West End
Llanelli,
Carmarthenshire
SA15 3DN

www.KSharp.co.uk
Twitter: @KSharpLtd
Email: Hello@KSharp.co.uk

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